Homework 3 — Results Report
Coffee Demand, Conduct, and Counterfactuals

Empirical Industrial Organization, Spring 2026

July 12, 2026

Data and settings. Annual panel, **52 countries, 1990–2019** (30 years, 35–50 active exporters per year, 1,334 country–year rows). Leaders: Brazil, Viet Nam, Colombia, Indonesia (the four largest exporters). Cooperative bloc: Peru, Honduras, Mexico. Supply and counterfactuals use **spec 2** (temperature-instrumented linear inverse demand). Counterfactual 1 uses **partial internalization** $\kappa = 0.5$ (not a merger). Negative equilibrium quantities trigger the **shut-down rule** (set $q = 0$, drop the country, re-solve). Marginal costs are recovered from the fitted demand curve, so the baseline Stackelberg equilibrium reproduces observed quantities exactly (zero baseline shutdowns); this isolates each counterfactual's effect. All monetary figures are in the price×quantity units of the data.

1 Demand estimation

Estimates of $P = \alpha_0 + \alpha_Q Q + \alpha_X X + \alpha_Z Z$ (and its log-log analogue), with $X =$ importer GDP, $Z =$ tea price. Instruments: fertilizer and temperature (spec 1) or temperature only (spec 2, used downstream).

Specification	α_Q (coef. on Q)	α_X	α_Z
Linear, OLS	−4.896	204.57	0.620
Linear, IV (fert+temp)	−3.778	156.38	0.582
Linear, IV (temp only, **spec 2**)	−0.927	33.53	0.487
Log-log, OLS	−3.305	6.63	1.527
Log-log, IV (fert+temp)	−3.756	7.54	1.574
Log-log, IV (temp only)	−11.006	22.16	2.326

All specifications give the expected signs: $\alpha_Q < 0$, $\alpha_X > 0$ (richer importers raise price at given quantity), $\alpha_Z > 0$ (higher tea price shifts buyers toward coffee). Spec 2 implies $b = -\alpha_Q = \mathbf{0.927}$ and a mean price elasticity of $\mathbf{-1.66}$ (range -3.79 to -0.81; see Figure 1), i.e. demand is in the mildly elastic region over most of the sample.

Important caveat (weak instruments). The demand slope is only weakly identified. Importer GDP alone explains 90% of the variation in world quantity, and adding the temperature instruments raises the first-stage R^2 only from 0.907 to 0.907: the *partial* first-stage F for the excluded instruments is **0.05**, far below any usable threshold. Consequently the spec-2 point estimate is imprecise and the specifications above disagree widely (α_Q from -0.93 to -4.9). We use spec 2 as instructed, but read the demand *level* with caution. This does not undermine the conduct and counterfactual conclusions below: those are driven by market *structure* (the number of followers) and by *ratios* of quantities, which are invariant to b; only the profit and surplus *levels* scale with b.

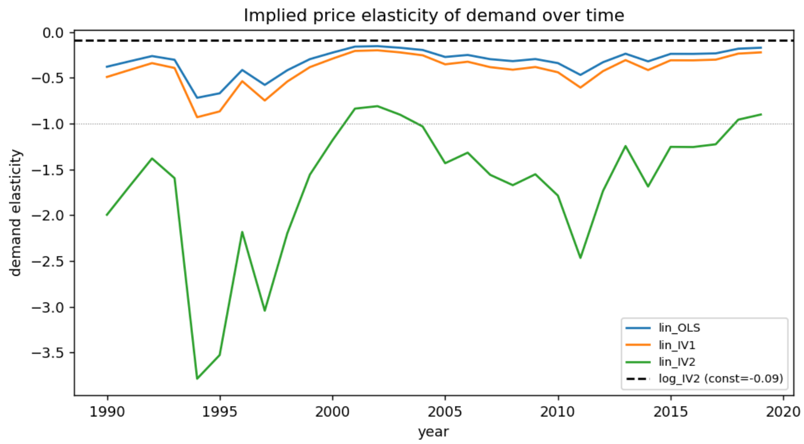

Figure 1: Implied price elasticity of demand over time (linear specs; log-log spec shown as a constant reference).

2 Conduct: marginal costs and markups

Recovered costs evaluate the conduct FOCs at observed prices and quantities. Under Cournot every country sets $c_i = P - bq_i$. Under Stackelberg, followers coincide with Cournot, while each leader's markup is scaled by the fringe pass-through $\tau = 1/(1 + N_F)$. With on average $N_F \approx 40$ active followers, $\boldsymbol{\tau \approx 0.024}$.

Leader	mean share of Q	Cournot markup	Cournot Lerner	Stackelberg markup	Stackelberg Lerner
Brazil	26%	21.9	0.187	0.55	0.0047
Viet Nam	16%	13.4	0.113	0.35	0.0029
Colombia	11%	9.6	0.082	0.23	0.0020
Indonesia	7%	5.7	0.048	0.14	0.0012

Difference between the two conduct assumptions (required discussion). Only the four leaders differ; every follower's recovered cost is identical across Cournot and Stackelberg. Because the fringe is large, $\tau \approx 0.024$, so the Stackelberg model attributes leaders a markup roughly $1/\tau \approx 41$ times smaller than Cournot does—Lerner indices collapse from 5–19% to 0.1–0.5%, and recovered marginal costs rise to just below the price. Economically: with $\sim$40 followers accommodating any leader output change, a leader's residual demand is nearly flat, its first-mover advantage is almost worthless, and the same observed exports are rationalized only by a cost close to price. The four leaders jointly supply $\approx$ 61% of world output yet, under Stackelberg, are modelled as essentially price-takers. This near-zero-margin feature drives the counterfactuals below. Per-leader figures:

3 Counterfactuals

Baseline Stackelberg equilibrium (year-averaged): $P = 132.9$, $Q = 89.9$, $CS = 3858$, leader profit $= 24.9$, follower profit $= 86.3$, total surplus $= 3969$, 0 shut-downs. Each table below is year-averaged.

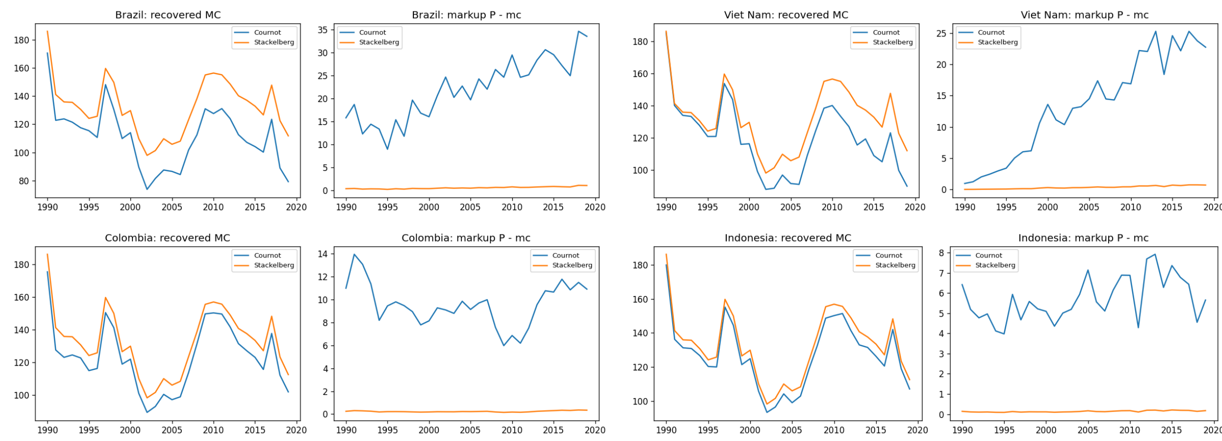

Figure 2: Recovered marginal cost (left panel of each pair) and markup (right) under Cournot vs. Stackelberg, by leader.

CF1 — Value of follower cooperation ($\kappa = 0.5$)

	baseline	CF1	Δ
Price	132.90	132.93	$+0.03$
Quantity	89.94	89.90	-0.03
Consumer surplus	3858.0	3855.1	-3.0
Leader profit	24.85	27.49	$+2.64$
Follower profit	86.33	77.91	-8.42
Total surplus	3969.2	3960.5	-8.7
Shut-downs	0	0.9	$+0.9$

Bloc gain $\Delta\Pi^{PHM} = -\mathbf{10.07}$ per year (summed over the sample, -302). Individual changes:

	base Π	coop Π	$\Delta\Pi$	% of base
Peru	8.45	4.08	-4.37	-66.3%
Honduras	13.30	9.93	-3.37	-47.7%
Mexico	7.27	4.93	-2.34	-59.5%

Answer. The value of cooperation is *negative*: all three members lose, so the maximum they would jointly pay to coordinate is zero—they would rather not. This is the Salant–Switzer–Reynolds merger paradox. A three-country bloc is far too small (a few percent of world output) to move the price when it restricts output (P rises by only 0.03), so internalization just makes bloc members cut sales while the ~41 outsiders—especially the leaders, whose profit rises +2.6—free-ride on the negligible price increase. On the "who would pay more as a percentage of base profit" question: none has a positive willingness to pay; Honduras is hurt least in percentage terms (-48%), Peru most (-66%). (Under a full merger, $\kappa = 1$, the loss is larger and the $11' + \Omega^F$ block is singular—partial internalization is what keeps the bloc interior; see the solution note.)

CF2 — Bad crop among leaders (+5% leader marginal cost)

	baseline	CF2	Δ
Price	132.90	134.17	+1.27
Quantity	89.94	88.57	−1.37
Consumer surplus	3858.0	3737.4	−120.7
Leader profit	24.85	0.00	−24.85
Follower profit	86.33	250.84	+164.50
Total surplus	3969.2	3988.2	+19.0
Shut-downs	0	4.0	+4.0

Leader quantity falls by 54.6 (all four leaders shut down); follower quantity rises by 53.2.
Answer: do followers benefit? Emphatically yes. A 5% cost increase exceeds the leaders' entire Stackelberg markup ($<$ 0.5% of price), so it pushes leader marginal cost above price and *all four leaders exit*. The competitive fringe absorbs almost exactly the vacated output (+53 vs. −55) at a higher price, and follower profit rises nearly seven-fold. This is the leader–follower structure in the extreme: the price increase the shock induces is a public good the fringe captures, and here the leaders are so marginal that they capture none of it themselves. *Caveat:* the total wipe-out is a knife-edge consequence of the near-zero Stackelberg leader margins (Section 2). Under Cournot conduct (margins 5–19%), the same shock would leave leaders active and the follower benefit would be modest—so this result is really a statement about the conduct assumption as much as about the shock.

CF3 — Tea becomes more competitive ($Z \to 0.95Z$)

	baseline	CF3	Δ
Price	132.90	132.89	−0.01
Quantity	89.94	82.78	−7.16
Consumer surplus	3858.0	3283.1	−574.9
Leader profit	24.85	21.80	−3.05 (−12.3%)
Follower profit	86.33	85.82	−0.51 (−0.6%)
Total surplus	3969.2	3390.8	−578.4
Shut-downs	0	4.4	+4.4

Answer: leaders lose more, both absolutely (−3.05 vs. −0.51) and in percentage terms (−12.3% vs. −0.6%). The effect works almost entirely through *quantity, not price*: the demand intercept falls, world quantity drops 7.2 (about 8%, with ∼4 marginal fringe producers exiting), while the equilibrium price barely moves (−0.01) because the output contraction offsets the demand shift. Leaders are hit harder because their thin margins make profit fragile to any squeeze, whereas the fringe holds most of the industry's profit and sheds only its marginal members.

CF4 — Value of leadership (Indonesia → follower)

	baseline	CF4	Δ
Price	132.90	132.92	+0.03
Quantity	89.94	89.91	−0.03
Consumer surplus	3858.0	3855.6	−2.4
Total surplus	3969.2	3971.2	+2.0
Shut-downs	0	0	0

Profit objects: $\Delta\Pi^{Indonesia} = -\mathbf{0.87}$ (leader 0.90 $\rightarrow$ follower 0.03); other leaders $+3.42$; original followers $+1.88$.

Answer. The strategic value of leadership is positive but small: Indonesia loses 0.87 when it gives up first-mover status ($\Delta\Pi^{Indonesia} < 0$). The magnitude is small precisely because leadership is nearly worthless in this model ($\tau \approx 0.024$). Notably, the *other* leaders gain the most ($+3.42$): with one fewer leader, Brazil, Viet Nam and Colombia face slightly less leader-side competition and their (tiny) rents rise, while the fringe also gains ($+1.88$). Consumer surplus dips marginally and total surplus edges up.

Shut-downs across counterfactuals

Baseline 0; CF1 0.9; CF2 4.0 (the leaders); CF3 4.4 (marginal fringe); CF4 0 per year. Because exit is a discrete margin, some responses are non-monotone (a price rise can pull a shut country back in). Country-level profit changes across all four counterfactuals:

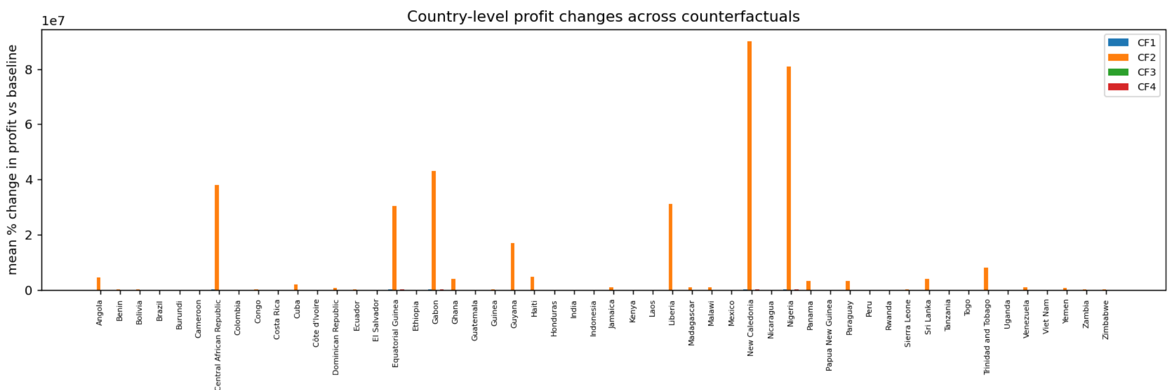

Figure 3: Mean % change in country-level profit vs. baseline, by counterfactual.

4 Summary

1. Demand slopes down with the expected signs, but the temperature instruments are weak conditional on importer GDP (partial $F = 0.05$); the demand level is imprecise, though the conduct/counterfactual *signs* are structural and robust to it.

2. The Stackelberg model assigns leaders near-zero markups ($\tau \approx 0.024$), in sharp contrast to Cournot (5–19%). This single feature explains every counterfactual.

3. Small-bloc cooperation *destroys* value ($\Delta\Pi^{PHM} < 0$): the merger paradox.

4. A modest leader cost shock eliminates the leaders and hands the market to the fringe—followers benefit enormously, but this is knife-edge given the thin leader margins.

5. A tea-driven demand contraction hurts leaders far more than followers, and operates through quantity (fringe exit), not price.

6. Leadership has a small positive strategic value; giving it up costs Indonesia 0.87 while helping the remaining leaders more.